3.6 Conversion Models Teacher Key

Teacher use

1. Table: 100, 200, 50, 250 centimeters. A centimeter is 1/100 of a meter, so 100 times as many units measure the same length.

2. 200 + 50 = 250. The second row is 2 hundreds, 5 tens, 0 ones. The digit 5 represented five tenths before multiplying and five tens afterward. Its value is 100 times as much. Distinguish the changed numerical value from the unchanged physical length.

3. 2.5 × 3 = 7.5 feet; 7.5 × 12 = 90 inches. Backcheck: 90 ÷ 12 ÷ 3 = 2.5 yards.

Feedback and retry

If a student says “move the decimal,” ask which unit became smaller and how many new units fit in one old unit. Then retry 0.75 meter = 75 centimeters. Use place-value reasoning before shortening to a calculation rule.